

"Physics as Information Processing" ~ Chris Fields ~

Lecture 2

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foreign it's June 15 2023 we are here in lecture two of Chris Fields course physics as information processing so thank you Chris looking forward to the lecture oh thank you Daniel um and yes welcome to this session uh this section is titled why quantum physics and if you will recall from the first session we reviewed the history of physics and and some of the history of mathematics and computer science or I'll the end from the end of the 19th century through to the beginning of the 21st Century and discuss the the slow development from classical thermodynamics of quantum information Theory and uh specifically characterizing Quantum information Theory as a new kind of physics that describes systems that are exchanging finite amounts of discreetly encoded information across some intervening boundary and so we use this very conventional graphical notation of two agents or physical systems Alice and Bob A and B that are exchanging energy and across some boundary and I always use a blue ellipse like this to indicate the boundary that separates Alice from Bob and this new approach to physics is entirely topological it's not geometric so it doesn't assume a particular background space time so it doesn't assume that Alice and Bob are spatially separated and so this this makes it a very different kind of theory from classical information Theory in that the channel uh that separates Alice from Bob this boundary is a boundary in state space it's not a boundary in some geometric embedding space and in particular it's not a boundary or a channel that's embedded in a three-dimensional space that separates Alice from Bob that's what we talked about last time and today what I want to discuss is how quantum theory in particular makes this idea of physics is a theory of communication simple than obvious and quantum theory of course has a terrible reputation uh of being abstruse and mathematically incredibly complicated and counter-intuitive and difficult to understand and this is one quotation among many of from leading physicists pointing out that that quantum mechanics is just difficult and uh as as you probably know there's an entire philosophical industry of interpretations of quantum theory that try to make sense of its ontology and so

what I don't want to do today is try to introduce quantum mechanics uh we're not doing uh any of this some of you will recognize this as the table of contents the first part of the table of contents of the famous textbook by Landau and Lifshits oh but if you've studied quantum theory in undergraduate or graduate school you've probably dealt with a textbook structured much like this one and you've probably been introduced to quantum mechanics as it was traditionally conceived as essentially a mechanical Theory a theory of motion uh with wave functions and particle representations and on and on and on so we're not going to try to do that this uh what we're going to do instead is take a completely information theoretic approach uh and we're going to characterize information transfer in a Quantum theoretic way without any assumptions about mechanics or space-time or any of so instead of this we're going to ask a simple question uh a simple by a simple question I mean a question with just a yes no answer or a binary answer so let's use up or down as our example question and to ask this question we need three things first we need an action we need a way of asking it uh for example making a sound or writing something down um second we need to a thing that we can ask the question of uh we need an environment or a friend or the rest of the world or an experimental apparatus or some system or other that we're going to act on to ask this question and the third thing that we need to ask this question is a shared language and it's the shared language that's often neglected in models based in physics um and I think the importance of the shared language was became clear and classical information Theory and it's it's become clear again of course in Quantum information Theory but this has only happened over the last few decades so what I want to do today is to construct in a step-by-step fashion a formal representation of these three things that we need to ask a simple question and what we'll find is that that formal representation is in fact quantum theory so it's quantum theory that provides us with a formal representation of asking these simple binary questions and of course if you can ask a binary question you can ask any finite number of binary questions so you can construct an arbitrarily complex finite decision tree that ends up formulating an arbitrary an arbitrarily complex but still finite question so this really is a general model of communicative interactions between agents so let's go about this and talk first about the action of asking a question and so the first thing I want you to do is actually say out loud up or down actually ask a question and then note two things about what just happened uh the first thing to note is that it took some time to do that that that action actually we required to find that amount of time um a couple of seconds not very long but it's not zero it always takes time to ask a question and the second thing to note is that it took energy to actually say up or down it

requires a little bit of exertion to ask a question so let's put those two together and Define a word we're going to define the word action to mean some energy that's spent in some time so asking a question is an action and it qualifies as an action because it requires spending some energy in some finite amount of time and so we can write this as action equals energy times time or I'll sometimes use the abbreviation E times t action is is always required when when there's some interaction or two-way action between systems so whenever we talk about physical interaction we're always talking about two systems doing something to each other that requires both energy and time so let's define another concept a concept of a minimum action needed to ask a question and specifically a question with a binary answer and we'll just pick a symbol to stand for that minimum action and the symbol we'll use because it's conventional as h and it turns out that this h is Planck's constant the number that Planck came up with when he was trying to solve the black body problem back in 1900 that we talked about last time and recall that was the problem of characterizing the spectrum of radiation emitted by some physical system that was hot how much energy comes out as a function of temperature and if you assume that the spectrum of energy coming out is continuous you get predictions that are wrong and plank got predictions that were correct by assuming that that Spectrum was discrete and that that this h uh represented the minimum amount of energy that could be emitted in a particular time so it represented the minimum energy times time or action so let's look at an example of that and here's an example that may be familiar it's the example of your own visual system so in the retinal the cells of your retinas uh there are many many molecules that are rhodopsins of one variety or other that respond to different wavelengths of visible light and they have some efficiency for capturing light and let's just assume that they're optimally efficient that there is efficient as they could possibly be and that means that they require the minimum amount of energy to change shape which is what they do when they're impacted by light to capture the the energy and hence the information that's available in the light and the minimum energy to do anything that records one bit of information so yes or no up or down uh is 0.7 so $\log 2$ of Boltzmann's constant times the temperature and again we saw this last time uh this is from roughly 1875. and was then built into classical by Shannon and then by landauer uh in the mid 20th century so what we're doing is is assuming that since rhodopsin's optimally efficient all it requires is this amount of energy to change shape and it's known experimentally that rhodopsin takes about 200 femtoseconds to change shape so that's 200 times 10 to the minus 15 seconds so it's a very very fast reaction chemical reaction so we can just calculate what the action is and at 37 degrees Centigrade so human body temperature KT

is about 4.3×10^{-21} Joule my joules just the standard unit of energy and so we can multiply everything together kT times 0.7 times this response time of 200 femtoseconds of multiplied together gives you an action of about 6×10^{-34} Joule seconds and if you're familiar with quantum mechanics you'll instantly recognize that number because it's very close to the value of Planck's constant uh which is 6.6×10^{-34} Joule seconds so this says that rhodopsin is very near to Optimal um the the best it would be able to do would be Planck's constant that's the minimum action that can be undertaken um so rhodopsin is clearly doing very well it can't be as efficient as we're assuming it to be uh but it's pretty efficient so this number Planck's constant is and this idea of a minimum action is not something we just pulled out of the air it's something that one can observe when you're looking at what real biological systems in particular do in other physical systems too so let's think about the the second component of of this project of asking a question and that's the thing to be asked and we want to ask a very simple question we want to have a theory of asking very simple questions so let's define the simplest thing that we can ask a binary question too and let's say this is the thing that we can ask a binary question of using just this minimum action $\hbar$ so we're going to ask for the minimal thing that it's possible physically to ask a question of and this thing needs a name so let's call it a qubit and that word is a contraction or Quantum bit and we can draw it like this a little sphere with an arrow through it Ah that's called a block sphere and we use this symbol q and that notation with the bar and the angle bracket uh is called a bra notation and it originates with Dirac so it's a traditional way of writing inner products actually in quantum theory [Music] to indicate the state of Q so we'll we'll this this thing called a qubit uh only responds to Binary questions so it clearly can have only two states the state that answers yes and the state that answers no and conventionally we we draw the arrow going up if the answer is yes and the arrow going down if the answer is no and so those are the two states that the qubit can be in when it's answered a question so now let's draw a picture for the act of asking a question of Q in the state q and let's draw it like this uh this curviero is the action of asking the question and getting the answer back and we're asking it of this minimum system the qubit and let's call this this entire Act of H and right at H acting on Q by this expression HQ so again we're developing the the notation of quantum theory which uh all of you who know the theory already will find familiar so the third thing that we have to to build into this picture and into the theory is the shared language so we have to know uh what we mean by up or down if we're going to ask the question up or down and we have to I'll have some representation of of what it means for the qubit to

distinguish up or down so let's say that to us oh up and down mean this little diagram with an arrow that's labeled up on one end and down on the other end so we're defining up and down in a very simple way as just opposites we don't know anything else about them except that they're opposites so this is a stripped down one bit question we could have called them one and minus one we could have called them A and B we could have called them anything uh we're just indicating that up and down are two opposing Concepts and so this object that we've used to make this definition of the relationship between our two words up and down is called a reference frame uh it's it's a a physical object that encodes a concept or a meaning and if you think about using language using English for example you can understand me speaking in English uh because your brain has the right sorts of neural circuits to allow you to understand English and produce English uh using your your hand in your mouth and those neural circuits are reference frames in your brain uh for understanding and producing English and they encode the relationships between different words so you all know about chat GPT you can think of all of the weights on uh all of the 10 to the 19th or however many parameters there are in gpt4 as a reference frame that describes the relationships between words and phrases in English and many other languages so this idea of a reference frame is an instance of of Land hours principle which is is often summarized as saying that information is physical what that means is that information always has to have some encoding and that encoding that object can always be thought of as a reference frame so let's put all this stuff together what we have now is a reference frame that defines the difference between up and down we have an action that we've labeled h and the action acts on this object of the qubit uh which is in some state and what we've drawn here now is an experiment and you can think of the incoming part of the arrow that goes from the reference frame to the qubit as a preparation of the qubit so you can think of it as I'm doing something to the qubit to put it in a particular State like up because that's the state I want it to be in and I can think of this returning part of the arrow as a measurement of the state that the qubit is actually in so if the qubit's actually in this state with Europe pointing up uh then I'm going to get back the answer up when I act on the qubit to measure its state so now we can recognize this this action h as the hamiltonian operator in quantum which always takes some time which we represent by an interval DT to act on some State and in particular this binary state q so this is this whole picture depicts the action of a very simple hamiltonian operator um and again I want to emphasize that we've just defined the qubit as the simplest thing that we can ask a binary question of so how does this H work uh we can use h to prepare this qubit in one of two states which we're going to label one and minus

one of the state one uh corresponds to the up Direction and the State minus 1 corresponds to the down Direction doing this to Q so preparing Q in one of these two states takes energy and we know how much uh boltzmann told us back in 1875 that the minimum energy to reduce the the entropy of the universe by one bit is $\log 2$ of KT we saw that earlier when we were thinking about rhodopsin so we can say that acting with h always requires some number times KT where that number is bigger than the log of 2. and remember the log of 2 is just in there to get units of bits of binary questions uh as opposed to units of gnats which are questions that have e answers where e is the exponential base the natural exponential base but we want to use units of bits so we have to always have this factor of $\log 2$. so this lets us write H in a particular way we can take out this this energetic term that vaultzmann tells us we have to include uh β times KT and what's left over is is in a sense the pure representation of the action stripping out the energy requirement and we're going to call that M because m is the standard symbol to use in quantum theory for a measurement operator and M is is now dimensionless H uh has units of energy KT is in energy and M has no units at all so it's h without the energy so now we can ask what is this operator M that we've defined as the the simplest possible action on a qubit but abstracting away the amount of energy that we have to use to act on the qubit well what M does is just encode a bit or decode a bit so we can use this operator M to say uh I want Q to have the value 1 I want it to be up so I can sort of reach out with my operator M and grab q and turn it so that its arrow is pointing up and that's what the the operation that's called preparation in physics it's what it means to prepare an experimental apparatus for example you you actually grab hold of the apparatus and you twist some knobs and flick some switches to prepare it to be in some state and you can use M to measure a state that isn't known so I can use M to grab a hold of Q and look at it and determine whether it's in state 1 or -1 and I can represent these as just a cycle that starts with measuring and then prepares or that starts with preparing and then measuring and if I represent it that way uh what I see is is m having a value which is either the value I want to encode or the value that I have measured and Q and M acts on that value and Q to produce a particular state of Q to put Q in a particular state or to realize that Q is in that state and so m is actually an identity acting on some value and Q so it Maps one and Q to 1 and Q and it Maps minus 1 and Q to minus 1 and Q if you do the whole cycle of preparation and then measurement and that makes sense it's actually an axiom in some formulations of quantum theory that if I measure the state of some object what I'm doing is also preparing it in that state so if I immediately measure it again I'll see the same value and that makes sense physically if I you know put my coffee cup on the

table and then I look for it it's likely going to be on the table unless something else intervenes well what does this mean it means that we can do a little bit of algebra it means that we can write this original expression of acting on Q with our hamiltonian in this somewhat more explicit form of using the energy $\beta K T$ and our operator M on Q and what I get out of that is the energy $\beta K T$ multiplied by Q so HQ equals EQ that's the Schrodinger equation so that's one of the central equations of quantum theory what it says is that this operator h is the total energy operator on a system its eigenvalues are the total energies of the system and we can now see where that energy is coming from in this very very simple strip down kind of representation of an action the energy is the thermodynamic energy that has to be spent to determine one binary outcome up or down now if you're familiar with quantum mechanics uh you will have seen exponentiated operators uh like the the next line here of algebra this operator P which is time dependent and which is the exponential of H uh multiplied by $i t$ over $\hbar$ is called the propagator it's the unitary operator in quantum theory the operator that conserves probability or conserves information and so using the representation of H as $\beta K T$ we can see that that this operator p is just e to the minus βm uh if we recall the wick rotation from the first session of this strange equivalence of $i t$ over $\hbar$ with 1 over $K T$ that wasn't discovered until 50 years after the invention of quantum theory um and that's still a very debated uh thing I mean people worry about what this what this equation really means but one thing that it means in this setting where H has this particularly simple form is that the propagator is now just the exponential of a number multiplied by this very simple strip down operator encodes a bit or decodes a bit but we know something about the exponential propagator just by looking at its form if you're again familiar with these sorts of exponential operators you know that they're wave equations and we can make sense of that using the diagram up above what m is doing is just performing a cycle uh it's just mapping 1 and Q back to itself or minus 1 and Q back to itself so we can think of of M as just taking this object and flipping it back around until it's uh until it gets itself again and that kind of cycle can always be turned into a wave equation and e to the i over $\hbar$ times $h t$ is a canonical form of a wave equation where the the horizontal axis here on the wave is time or if you look at the little cycle that I've drawn as a circle with an arrow it's it's the time of the thing spinning around like this which is exactly the same thing as a wave traveling through time so that tells us something very important it tells us that that this equivalent expression e to the minus βm is also a wave equation so it tells us that this operator m has built into it some sense of time um it's the sense of time of this prepare measure cycle and that's implicit in m right we've used the wick

rotation to get rid of the the kind of what what I'll end up calling the external time T the objective time t and we'll see in July uh where this internal time comes from uh in terms of the wick rotation but we can say now that it's an agent specific time it's an internal time reference frame that's built into the our idea of this system Alice that can act on the world and it represents the time that's required to ask this question okay Chris uh can I ask a question at this point yes uh so on this slide uh with the operator U I'm curious that's how can the agent deploy in the operator m tell the difference or break the Symmetry between preparation and measurement or are they just completely dual pictures yeah they're completely dual so the agent the agent never breaks the Symmetry the agent can't actually distinguish the act of preparing from the act of measuring every preparation is a measurement and every measurement is a preparation right that's that's in fact what this Axiom of measuring a state leaves the particle in the measured State means right so yeah mathematically they're duals thank you okay so let's let's now use this theory that we've developed to describe the interaction between Alice and Bob and let's think of this situation in which Alice prepares a qubit that Bob then measures and then Bob prepares the very same qubit and that Alice measures it so Alice acts with her hamiltonian operator which will label H_A and Bob acts with his hamiltonian operator that we label H_B and let's say that Alice prepares the qubit as in-state up and uh Bob can then measure the qubit in state up and say oh the qubit is now up and if Bob then prepares the Cuban State down uh Alice can measure the qubit in state down and say oh the qubit switch from the up that I prepared to down so Bob must have prepared it is down so what's happened here uh Alice and Bob have actually shared a bit flip right Alice said uh up and and Bob said down so and Alice heard him say down so they've they've shared this idea of change by interacting with each other now this works as long as they mean the same thing by up and down so we've implicitly assumed in talking about them communicating that they both have this reference frame that distinguishes up from down and this is actually a completely General model of what I'll call a noiseless one qubit channel now what does that mean it's noiseless because there's nothing else in the picture right there's nothing else in the picture that can jiggle the qubit when else and Bob aren't looking Alice and Bob are the only systems we're talking about and they share this qubit and there's nothing else in the picture so there's no kind of third party or environment or heat bath or anything like that that's going to mess with the qubit and disrupt their interaction so that's a noiseless channel and it's a one qubit channel because they just have one qubit that they're interacting with and they're alternately preparing and measuring at state now clearly if if they can interact using one qubit they can interact

using any finite number of qubits so here's a picture of an in qubit Channel that uh Alice and Bob which here I've labeled A and B because I took this figure from a paper are sharing and uh Alice and Bob both have n operators one for each of the N qubits that they share and they just do this prepare and measurement cycle on all in qubits so you can think of Alice preparing the in qubits Bob measuring the in qubits and then Bob preparing an ALICE measuring and that allows them to to exchange in-bit messages and we can make n as large as we want and so they can exchange messages of arbitrary complexity and if they share the meaning of up and down then they can exchange these messages with no noise so they can communicate and we now have our theory of two communicating agents that are that are sharing arbitrary uh finite information now I've labeled uh q_1 through q_n in this picture uh a holographic screen and in fact that's exactly what it is I'll recall from Session One this slide illustrating the holographic principle Shannon made this point that a bit of information is an answer to a yes no so we can write down a number of possible States we can write down an entropy um and the limiting case of that entropy is the entropy of a black hole the densest possible uh encoding of information in space or on a boundary and this holographic principle allows us to see any boundary as we discuss in session one as an encoding of bits so the previous boundary of these in qubits we can think of as a holographic screen now why is that important it's important because it allows us to use this idea of holography and this idea of fixed encoding of a message or a fixed encoding of an observable entropy at some density that was developed in cosmology to talk about arbitrary Quantum communication channels so we can import a whole bunch of theory that was done someplace else into Quantum information Theory and see that it works perfectly because it describes exactly the same situation what does this tell us philosophically it tells us that Alice and Bob can learn about each other at most in bits per communication cycle so the maximal amount of information that they can get from an observation is the in bits that the other one wrote on the screen so this tells us something very important that when an agent is looking at their environment which is this other agent Bob the amount of information that can be extracted from the environment is actually proportional to the size in bits of the boundary and the holographic principle just makes that size finite in space and I said early on that there's no embedding space time in this theory that Alice and Bob are not thought of as separated in space but as soon as we use the holographic principle we can think of the boundary between them as having a spatial coordinate even though they don't have a spatial coordinate so uh we'll get back to this point in September when we talk about theories of emergent SpaceTime but all of these theories of emergent space-time come

back to this idea that boundaries can have spatial extent even though the systems that they separate don't have any natural spatial description or natural spatial degrees of freedom okay so now let's recall another picture from Session One we talked a little bit about feynman's theory of scattering and the the origin of Feynman diagrams and the idea that if Alice prepares some system in some initial state and then measures the state later if we want to calculate what the new state is we have to take into account all of the possible paths that join the initial state that Alice prepared to the final state that Ellis measured so what does that mean I mean here effectively if we think of suppose I'm Alice and the boundary is my I'm talking to the screen and then later on I'm going to measure and between my preparation actions and my measurement actions a bunch of stuff happens on the other side of the screen and if I want to predict what's going to happen on the other side of the screen I've got to take into account every possible thing that could happen and uh I have to give some weight to those possible things that could happen and quantum theory in fact tells you how to assign those weights to things that could happen in order to make a measurement given some theory of the mechanism that makes things happen but the fact that all of the paths have to be counted as what unitary Evolution means this in this this Evolution that preserves information that doesn't take any away it doesn't add any it just represents it in different ways so if you look at the screen and imagine that Alice is interacting with her boundary and then she's interacting with her boundary again it looks just like the previous screen and the way they read arrows are what's happening inside Bob the wavy red arrows are a representation of what's going on inside Bob as he measures his side of the screen and then prepares his side of the screen so what Feynman is representing here is just the action of the environment or the action of the system uh while Alice isn't looking between Alice's preparation stage and her measurement stage while Bob is is doing his measurement in preparation so this diagram is actually completely symmetrical we could have drawn an arrow where the the wavy red things are and drawn wavy red things where we've talked about Alice and we'd have the same picture just from Bob's point of view we can also think of this of course as Alice communicating with her first future self by writing a memory record someplace in the world and then coming back and reading it and if Alice writes a memory on the world and then comes back later on and reads it she actually has to take into account some theory of all of the things the world might have done to her memory while she wasn't looking um so this picture of feynman's actually tells us a lot about memory and when we think about organisms running around in the world uh doing like we do all the time um we tend to think of memories as things that are are persistent and unchanging and all

of that but we know that that's not true we know that memory is degraded in various ways uh we know that uh adversarial third parties can come and and change what we've done so that they fool us into thinking that we've remembered something that we haven't and so on and so forth that whole picture of memory is actually encoded in in this picture that Feynman drew in the late 50s to represent scattering of particles so what we've done here is started with this simple idea of asking questions and we've actually constructed a fair amount of quantum theory we constructed the Schrodinger equation we constructed the unitary propagator we constructed the idea of a reference frame which we'll talk about intensely next time in July so we can sum this up by saying that quantum theory simply is a theory of and that quantum mechanics of the theory that Feynman was complaining about in that earlier slide the quantum mechanics generates this problem that seems intractable of how to understand measurement because the theory is in fact about measurement it's not about mechanical motion at all so Quantum Theory actually tells us nothing about ontology and this is why the interpretations of quantum theory the the many worlds interpretation where you have all these branching universes uh and so forth are not really science they're philosophy their attempts to understand an ontology that's generated by a theory that doesn't talk about ontology it talks about communication between agents but we've left something very important out we've left out a consideration of what happens when Alice and Bob don't mean the same thing by up and down and we've left out the famous phenomenon of entanglement spooky action at a distance as Einstein called it and it turns out these things that we've left out are very very closely related and we're going to see why in July and I'm going to give you a brief preview now so what happens when Alice and Bob actually mean different things by up and down what if their reference frames aren't aligned well the first thing to note is that nothing in the theory says they have to be aligned so we have to consider the possibility that they're not and in fact both Alice and Bob have to have what's called free choice of reference frame when Alice prepares her her instrument she has to be able to prepare it any way she wants to in order to do an experiment and uh when Bob makes up a sentence to tell Alice uh he has to be able to choose any language he wants to use he's a he's a free agent in this way and if they don't have that freedom it turns out they're entangled so reference frame non-alignment and entanglement are very very closely coupled now misalignment of reference frames is what produces superpositions of outcomes so if Alice makes her measurements with her up and down arrow arranged vertically for example but Bob prepares his qubit with his up and down Arrow arranged at 90 degrees to Alice's Alice is not going to have any idea what Bob did I mean she

can't see distinctions in the left right direction she can only see distinctions in the up and down Direction so all she can do is flip a coin if she wants to know what uh Bob prepared it's 50 percent likely he prepared up for her what she calls one and it's fifty percent likely that he prepared what's down for her she can't tell oh now if Bob if Bob's reference frame is pointing at about 45 degrees so up for him means 45 degrees off of Alice's up then she can sort of tell what he did if he's pointing 45 degrees from Up then uh she's probably going to measure up and if he's putting 45 degrees from town she's probably going to measure down so her outcomes are 75 percent for one and 25 percent for minus one if he prepares uh at this 45 degree angle so we can write this in a particular way we can say that the value that Alice gets acting with her M on a qubit that Bob prepared is some number times what she measures is up plus some other number plus times what she measures as minus one where those numbers the squares of those numbers add to one so why do they have to add to one well they add to one because of the Pythagorean theorem one is the distance the total distance between the two outcomes $a^2 + b^2$ are the other two sides of that triangle so another way to think of it is that the metric distance impossibility space is one and a^2 and b^2 are just the euclidean distance components again euclidean distance is based on the so this is the born rule in quantum uh it's the rule that tells us what happens when we're trying to measure a superposition and it tells us where these superpositions come from uh we have a mismatch in in reference frames between two agents that are communicating if we go back to the free energy principle which is what uh this whole active inference class is about and we formulated in quantum theory we get a very simple idea of what the free energy principle is the fep and quantum theory just says that interacting systems are going to behave in a way that aligns their reference frames and that makes their communication as good as it can be uh but since aligning reference frames perfectly leads to entanglement of the free energy principle drives systems asymptotically toward entanglement well the principle of unitarity says that all systems will approach entanglement asymptotically so we can see the fep is actually a classical statement of the principle of unitarity and what we'll see in the next session is why this is the case why active inference when we think of a Quantum mechanically is a way of talking about the principle of unitarity which is the deepest Axiom of quantum theory and we'll also see that this question of alignment becomes more and more complex as the complexity of the messages of the CIS that the systems are trying to communicate to each other increases and the message complexity correlates with system complexity so that aligning complicated systems is very hard although aligning very simple systems is fairly easy if you

think about uh what's the what's the simplest theory in science it's particle physics because you're dealing with these very simple things that interact in extremely simple ways and the really difficult parts of science are things like sociology where you're dealing with uh incredibly complicated systems like humans that interact in incredibly complicated ways and all we have are very vague kind of folk theories of how that works but we know from that example that aligning complex systems is extremely it's what humans don't get along very well so the second session uh well the second discussion section which Andrew will be leading is on Saturday the first of July a couple of weeks from now the third lecture session will be on July 13th and uh I hope you all will contribute to the interactive question and answer session on the web and uh show up for the discussion on Saturday the 1st of July and join us again in July for the discussion of quantum reference frames and reference Frame Alignment so thank you very much thank you Chris all right thank you all again please check out the chorus syllabus ask questions through the site and see you at the discussion see you next thank you bye